Similar questions

Five new questions practising the same skills as the exam question above. They are not past exam questions. Marks are as in the original. Answers are at the end.

1 Express $\dfrac{a^{\frac{9}{2}} - a^{\frac{7}{2}}}{a^{\frac{3}{2}} - a}$ in the form $a^m + \sqrt{a^n}$, where m and n are integers and $a \neq 0$ or 1. **[5]**

2 Express $\dfrac{a^5 - a^4}{a^{\frac{5}{2}} - a^2}$ in the form $a^m + \sqrt{a^n}$, where m and n are integers and $a \neq 0$ or 1. **[5]**

3 Express $\dfrac{a^{\frac{5}{2}} - a^{\frac{3}{2}}}{a - a^{\frac{1}{2}}}$ in the form $a^m + \sqrt{a^n}$, where m and n are integers and $a \neq 0$ or 1. **[5]**

4 Express $\dfrac{a^{\frac{9}{2}} - a^{\frac{7}{2}}}{a - a^{\frac{1}{2}}}$ in the form $a^m + \sqrt{a^n}$, where m and n are integers and $a \neq 0$ or 1. **[5]**

5 Express $\dfrac{a^6 - a^5}{a^{\frac{3}{2}} - a}$ in the form $a^m + \sqrt{a^n}$, where m and n are integers and $a \neq 0$ or 1. **[5]**

Question 1

numerator $= a^{7/2}(a-1) = a^{7/2}(\sqrt{a}-1)(\sqrt{a}+1)$; denominator $= a(\sqrt{a}-1)$. So the expression is $a^{5/2}(\sqrt{a}+1) = a^3 + a^{5/2} = a^3 + \sqrt{a^5}$

Question 2

numerator $= a^4(\sqrt{a}-1)(\sqrt{a}+1)$; denominator $= a^2(\sqrt{a}-1)$. So the expression is $a^2(\sqrt{a}+1) = a^{5/2} + a^2 = a^2 + \sqrt{a^5}$

Question 3

numerator $= a^{3/2}(\sqrt{a}-1)(\sqrt{a}+1)$; denominator $= a^{1/2}(\sqrt{a}-1)$. So the expression is $a(\sqrt{a}+1) = a^{3/2} + a = a + \sqrt{a^3}$

Question 4

numerator $= a^{7/2}(\sqrt{a}-1)(\sqrt{a}+1)$; denominator $= a^{1/2}(\sqrt{a}-1)$. So the expression is $a^3(\sqrt{a}+1) = a^{7/2} + a^3 = a^3 + \sqrt{a^7}$

Question 5

numerator $= a^5(\sqrt{a}-1)(\sqrt{a}+1)$; denominator $= a(\sqrt{a}-1)$. So the expression is $a^4(\sqrt{a}+1) = a^{9/2} + a^4 = a^4 + \sqrt{a^9}$